

Vehicle Safety Intelligence

Text-Only Severity Prediction from NHTSA Vehicle Complaints

Deep Learning Final Project — Course 35374

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1. Introduction and Problem Definition

The objective of this project is to predict whether a vehicle complaint describes a serious safety failure using only the complaint's free-text description. The data comes from the U.S. National Highway Traffic Safety Administration (NHTSA) owner-complaint dataset. The practical motivation is complaint triage: a model that can rank or flag potentially serious reports could help reviewers focus attention on complaints that are more likely to involve crashes, fires, injuries, or deaths.

The task is formulated as binary classification. A complaint is labeled serious when at least one of four conditions is present: a reported crash, a reported fire, one or more injuries, or one or more deaths. Importantly, the fields used to construct this target are not used as model inputs. The models receive complaint text only.

The project compares three neural approaches with materially different text representations: a TF-IDF dense multilayer perceptron (MLP), a regularized one-dimensional convolutional neural network (CNN), and a bidirectional LSTM recurrent neural network (BiLSTM). All models use the same data split, class weighting strategy, and evaluation pipeline.

2. Dataset and Target Construction

The source file is the NHTSA FLAT_CMPL complaint dataset. The pipeline loads the documented 49 fields and is robust to occasional trailing or missing columns. Up to 300,000 rows are used, with stratified subsampling when the source contains more records.

2.1 Target construction

Four source columns are transformed to create the binary target:

- CRASH_b = 1 when CRASH is 'Y', otherwise 0.
- FIRE_b = 1 when FIRE is 'Y', otherwise 0.
- INJURED_NUM = numeric conversion of INJURED, with invalid or missing values replaced by 0.
- DEATHS_NUM = numeric conversion of DEATHS, with invalid or missing values replaced by 0.
- serious_failure = 1 when CRASH_b = 1 OR FIRE_b = 1 OR INJURED_NUM > 0 OR DEATHS_NUM > 0; otherwise 0.

In the held-out test set used for the supplied results, 1,495 of 22,605 complaints are positive, corresponding to a serious-failure rate of 6.61%. This substantial imbalance is important because a naive classifier could obtain high accuracy simply by predicting the majority class.

2.2 Train, validation, and test split

The split is performed by ODINO rather than by individual row. One incident may appear in multiple rows because different components can be associated with the same complaint. Grouping by the unique ODINO complaint identifier reduces the risk of near-duplicate complaint text appearing in both training and evaluation data. Unique ODINO values are randomly assigned to approximately 70% training, 15% validation, and 15% test partitions.

2.3 Class weighting

All three models use class weights during training. The negative class receives weight 1.0 and the positive class receives weight equal to the number of negative training examples divided by the number of positive examples. This increases the loss contribution of rare serious complaints and discourages the networks from minimizing loss mainly by favoring the majority class.

3. Text-Only Feature Engineering

The only predictive input is CDESCR, the free-text complaint description. Missing descriptions are replaced with an empty string. No vehicle make, model, manufacturer, VIN, state, component category, vehicle age, speed, or incident flag is supplied to the networks.

Two text representations are prepared. Model 1 uses TF-IDF with up to 3,000 features, unigrams and bigrams, a minimum document frequency of five, sublinear term frequency, and Unicode accent stripping. Models 2 and 3 use a learned sequence representation: TextVectorization creates sequences of 200 tokens from a vocabulary of up to 20,000 tokens, followed by a 64-dimensional trainable embedding.

COMP_TOP is derived from COMPDESC only for subgroup analysis after prediction. It is never used as an input feature. This distinction allows the project to examine performance by component without leaking the official component category into model training.

4. Model 1 — TF-IDF + Dense MLP

Model 1 is the baseline neural model. It receives the 3,000-dimensional TF-IDF representation and passes it through Dense(128, ReLU), Dense(64, ReLU), and a single sigmoid output unit. The model is trained with binary cross-entropy and Adam. EarlyStopping monitors validation loss with patience 2 and restores the best weights.

This architecture deliberately ignores word order. It tests how much of the severity signal can be captured by vocabulary and short n-gram frequency alone.

4.1 Test performance

On the serious_failure target, Model 1 achieves ROC-AUC 0.9428, accuracy 0.9399, precision 0.5310, recall 0.7860, F1 0.6338, and Brier score 0.0459 at threshold 0.5. Its strong AUC indicates that relatively simple lexical information already separates serious from non-serious complaints well.

5. Model 2 — Regularized 1D CNN

Model 2 uses token sequences and a learned embedding. Its architecture is Embedding(20,000, 64) → Dropout(0.3) → Conv1D(64 filters, kernel size 5, ReLU) → GlobalMaxPooling1D → Dense(32, ReLU) → Dropout(0.3) → Dense(64, ReLU) → sigmoid.

L2 regularization of 1e-4 is applied to convolutional and dense layers. Adam uses a reduced learning rate of 5e-4, and EarlyStopping monitors validation loss with patience 3. The CNN can learn local phrase

patterns that TF-IDF cannot represent as flexibly, while Global Max Pooling retains the strongest detected pattern across the complaint.

5.1 Test performance

Model 2 is the strongest uncalibrated model in the supplied run: ROC-AUC 0.9513, accuracy 0.9616, precision 0.6807, recall 0.7913, F1 0.7318, and Brier score 0.0291. It improves both ranking performance and the precision-recall balance relative to Model 1.

6. Model 3 — Bidirectional LSTM

Model 3 is a recurrent architecture intended to capture longer-range sequential dependencies. The network uses Embedding(20,000, 64) → SpatialDropout1D(0.3) → Bidirectional LSTM(48 units per direction, dropout 0.3) → Dense(32, ReLU) → Dropout(0.3) → Dense(64, ReLU) → sigmoid. Adam uses learning rate 5e-4 and EarlyStopping uses patience 3.

Unlike the CNN, which emphasizes local windows, the BiLSTM reads the sequence in both directions and can theoretically encode relationships over longer spans of text. In this dataset, however, the added sequential complexity does not improve performance.

6.1 Test performance

Model 3 obtains ROC-AUC 0.9337, accuracy 0.9506, precision 0.5953, recall 0.7920, F1 0.6797, and Brier score 0.0398. Recall is similar to the other models, but ranking and precision are weaker than the CNN.

7. Model Comparison

Model	AUC	Accuracy	Precision	Recall	F1	Brier
M1 TF-IDF MLP	0.9428	0.9399	0.5310	0.7860	0.6338	0.0459
M2 CNN	0.9513	0.9616	0.6807	0.7913	0.7318	0.0291
M3 BiLSTM	0.9337	0.9506	0.5953	0.7920	0.6797	0.0398

ROC-AUC, shown in Figure 1, is the primary comparison metric because the positive class is rare and accuracy alone can be misleading. Model 2 has the highest AUC and F1 and the lowest Brier score among the raw models. Bootstrap analysis gives 95% AUC intervals of [0.9349, 0.9500] for M1, [0.9443, 0.9578] for M2, and [0.9255, 0.9414] for M3. The intervals show that the CNN is consistently competitive, while the BiLSTM does not justify its additional sequential complexity in this run.

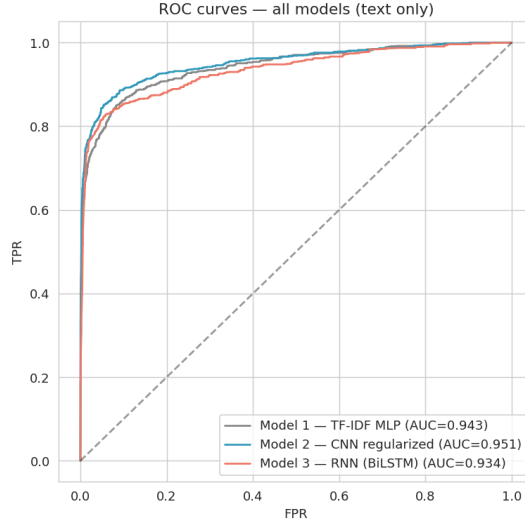


Figure 1. ROC curves for all three models on the held-out test set.

Precision-Recall curves are also important because they focus directly on performance for the minority positive class. The project therefore evaluates ROC and PR behavior rather than relying on accuracy alone.

8. Calibration and Decision Thresholds

After model comparison, the model with the highest test AUC in the implemented pipeline is selected for probability calibration. In the supplied run this is Model 2. Isotonic regression is fitted using validation predictions and validation labels, then applied to test predictions.

Calibration addresses a different question from discrimination. ROC-AUC measures whether the model ranks serious complaints above non-serious ones; calibration measures whether a score such as 0.70 behaves like an empirical probability of approximately 70%. The reliability diagram in Figure 2 compares raw sigmoid outputs with isotonic-calibrated probabilities.

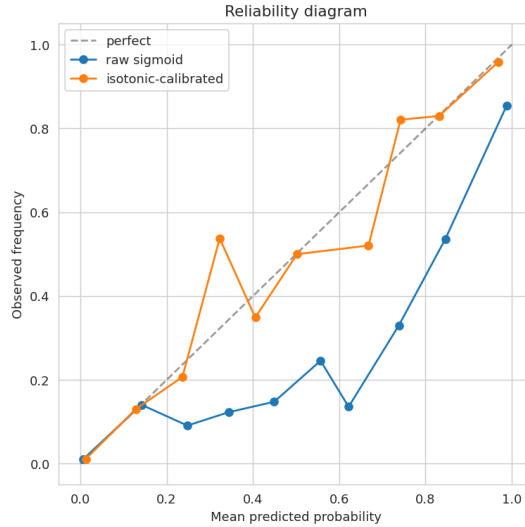


Figure 2. Reliability diagram for Model 2: raw sigmoid outputs versus isotonic-calibrated probabilities.

For Model 2, the Brier score improves from 0.0291 before calibration to 0.0232 after calibration. At a fixed 0.5 threshold, calibrated predictions have accuracy 0.9724, precision 0.8712, recall 0.6829, and F1 0.7657. This illustrates why threshold selection must reflect the application: increasing precision can reduce recall. In safety triage, a lower threshold may be preferable if missing serious complaints is more costly than reviewing additional false positives.

9. Error and Subgroup Analysis

The project evaluates the calibrated best model across component categories, even though component labels are not supplied during training. This provides a diagnostic view of where text-based severity detection is stronger or weaker.

Among component groups meeting the analysis sample requirements, performance varies materially, as shown in Figure 3. Examples include Suspension (AUC 0.854), Exterior Lighting (0.888), Power Train (0.917), Electrical System (0.934), Service Brakes (0.940), Vehicle Speed Control (0.968), Air Bags (0.975), and Service Brakes, Hydraulic (0.997). Some of these groups have relatively few positive examples, so subgroup results should be interpreted cautiously.

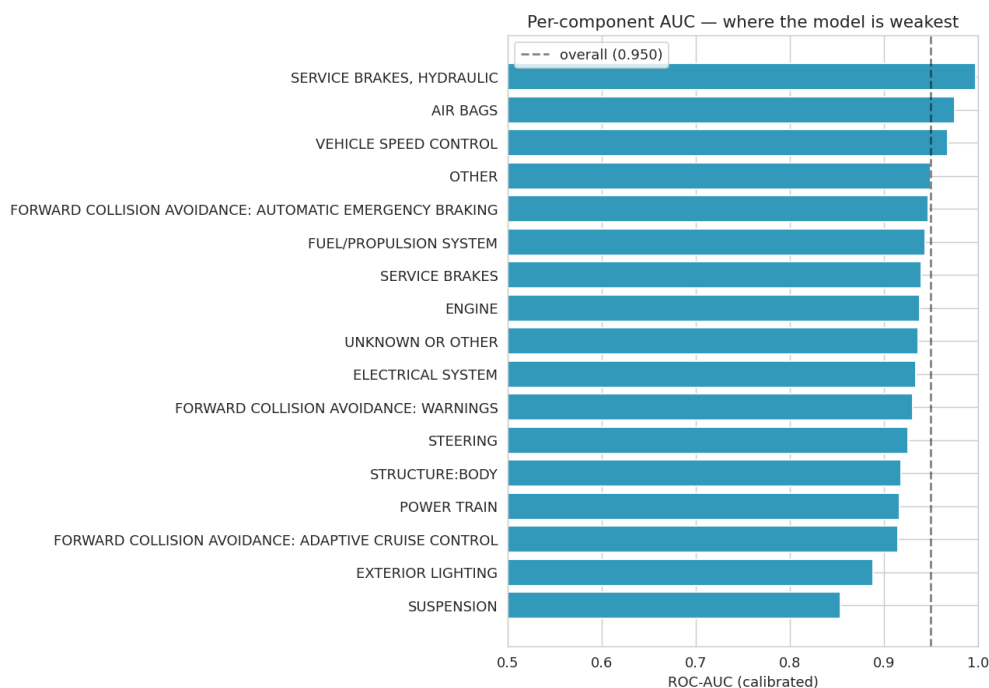


Figure 3. Calibrated ROC-AUC by component category, ordered from strongest to weakest.

The pipeline also extracts representative true positives, true negatives, borderline cases, false positives, and false negatives across the calibrated score range. These examples make model errors inspectable and help determine whether mistakes reflect ambiguous complaint language, missing context, or model limitations.

10. Interpretation and Discussion

The three models show that complaint text contains substantial information about event severity. The strong TF-IDF baseline suggests that severity is often signaled by explicit vocabulary and short phrases. The CNN improves on this by learning local phrase patterns directly from token sequences. The BiLSTM's weaker result suggests that longer-range sequence modeling is not automatically beneficial for this task.

A key strength of the design is the separation between label construction and model input. Crash, fire, injury, and death fields define the gold target but are withheld from training. Likewise, component information is used only for post-hoc analysis. The ODINO-level split further reduces leakage between train and test data.

The main limitation is selection bias: the model estimates severity among complaints that were submitted to NHTSA. It does not directly estimate the safety of all vehicles on the road. The data also reflects the wording and completeness of complainants' English-language descriptions. A high model score should therefore be interpreted as a triage signal, not as proof that a safety event occurred.

11. Recommended Model and Practical Use

For maximum predictive performance in the supplied experiment, Model 2 — the regularized CNN — is the recommended model. It has the highest raw ROC-AUC (0.9513), the strongest F1 score (0.7318), and the best raw Brier score (0.0291), and it supports useful post-hoc calibration.

Model 1 remains an attractive operational baseline because it is simpler and still achieves AUC 0.9428. If computational simplicity, interpretability of the text representation, or deployment cost dominates the small predictive gain, M1 could reasonably be preferred. The BiLSTM is not recommended here because it is more computationally demanding while producing the lowest AUC.

A realistic application would be a human-in-the-loop complaint triage system. New complaint text could receive a calibrated severity score and be prioritized for review. The decision threshold should be selected according to operational costs, particularly the cost of false negatives.

12. Future Work

- Tune the operating threshold using an explicit cost function rather than a default threshold of 0.5.
- Perform temporal validation to test whether performance remains stable on later complaint periods.
- Investigate transformer-based text encoders and compare their gains against computational cost.
- Analyze systematic false negatives and false positives by complaint type and writing style.
- Extend subgroup analysis to manufacturer and vehicle model for diagnostics, while keeping those fields out of model training.
- Evaluate calibration stability on new time periods and recalibrate when necessary.
- Explore component-aware early-warning aggregation at the vehicle make/model level.

13. Conclusion

This project demonstrates that serious vehicle-safety complaints can be identified with high discrimination using the complaint text alone. Three neural approaches were compared under a common ODINO-based split and class-weighted training procedure. The regularized CNN achieved the best test ROC-AUC at 0.9513, followed by the TF-IDF MLP at 0.9428 and the BiLSTM at 0.9337.

The results indicate that much of the predictive signal is lexical and local rather than dependent on long-range sequence structure. Calibration further improves the usefulness of model scores as probabilities, while threshold analysis exposes the trade-off between precision and recall. The resulting system is best viewed as a prioritization tool for human reviewers rather than an autonomous determination of vehicle safety.

Appendix A — Key Implementation Parameters

- Maximum sample size: 300,000 complaints.
- Split: approximately 70% train / 15% validation / 15% test by ODINO.
- TF-IDF: 3,000 maximum features; unigrams + bigrams; min_df = 5.
- Sequence vocabulary: 20,000 tokens; sequence length: 200.
- Embedding dimension: 64.
- Batch size: 512.
- M1: Dense 128 → Dense 64 → sigmoid; Adam; EarlyStopping patience 2.
- M2: Conv1D 64, kernel 5; Dropout 0.3; L2 = 1e-4; Adam LR 5e-4; EarlyStopping patience 3.
- M3: Bidirectional LSTM 48 units/direction; dropout 0.3; Adam LR 5e-4; EarlyStopping patience 3.
- Loss: binary cross-entropy; primary evaluation metric: ROC-AUC.
- Calibration: isotonic regression fitted on validation predictions.